OBJECT ORIENTED SOFTWARE ENGINEERING

Ms. Archana A

Department of Computer Applications

OBJECT ORIENTED SOFTWARE ENGINEERING

Procedural Sequence Models

Archana A
Department of Computer Applications

Procedural Sequence Model…

- Sequence diagrams contains independent object.

The UML has elaborations for sequence diagrams to show procedural calls

1. Sequence diagram with Passive objects.
2. Sequence diagram with Transient objects

1. Sequence diagram with Passive Objects

- With procedural code all objects are not constantly active.
- Most objects are passive and do not have their own threads of control.
- A passive object is not activated **until it has been called**.
- Once the execution of an operation completes the control returns to the caller, the passive objects become inactive.

1. Sequence diagram with Passive Objects…

For Example:

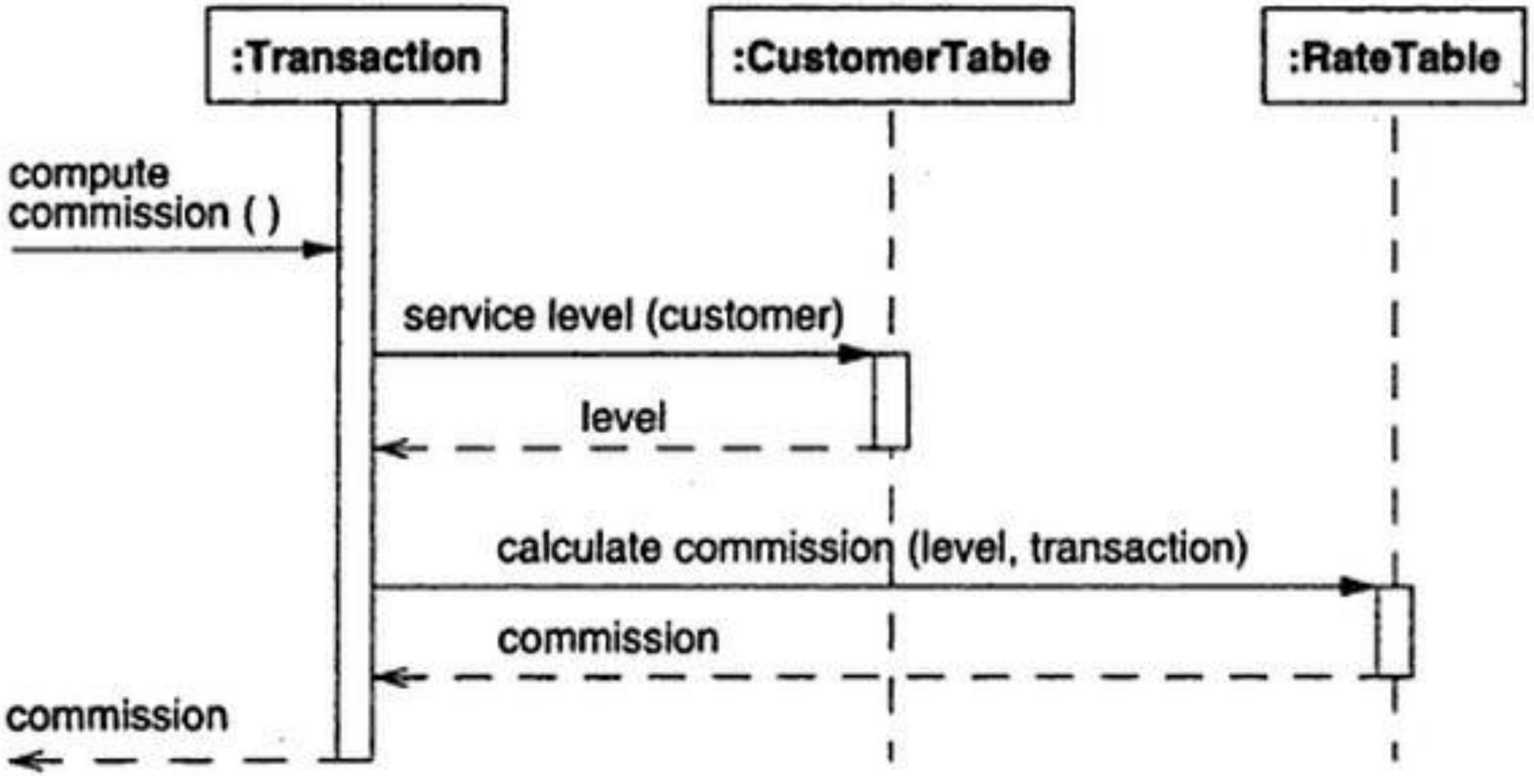


Figure 8.5 **Sequence diagram with passive objects.** Sequence diagrams can show the implementation of operations.

1. Sequence diagram with Passive Objects…

- The example calculates commission for stock brokerage transaction.
- The transaction objects receives a request to compute commission.
- It obtains customer's service level from customer table then asks the rate table to compute commission based on the service level, after which it returns commission value to the caller.

1. Sequence diagram with Passive Objects

In the given example

- Thin rectangle shows period of time for an object execution.
- This is called activation or focus of control.
- Dashed line shows period of time when an object exists but not active.
- The entire period during which object exists is called **life line.**

1. Sequence diagram with Passive Objects…

Example:

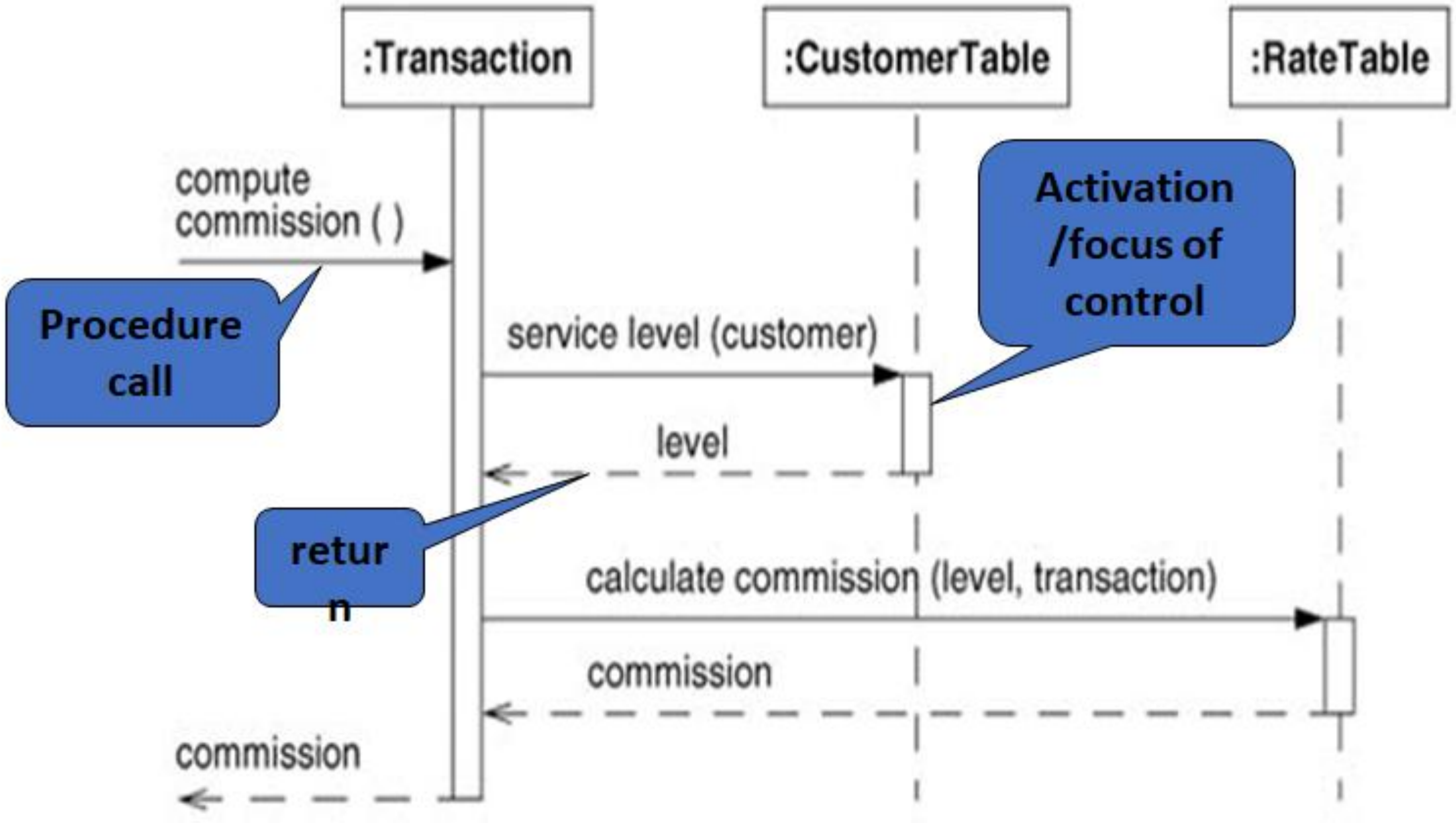
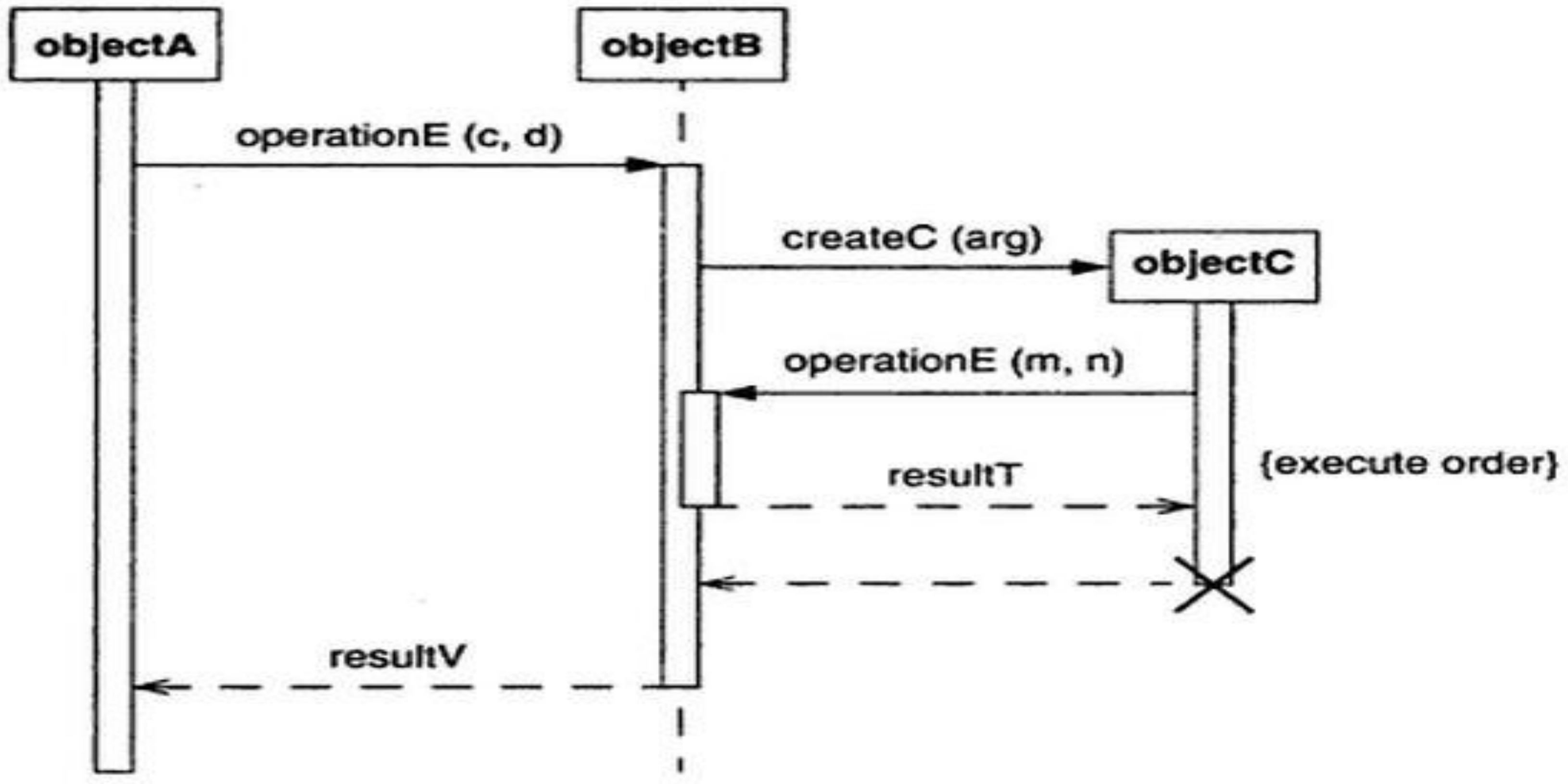


Figure **Sequence diagram with passive objects.** Sequence diagrams can show the implementation of operations.

2. Sequence diagram with Transient Object

Consider an example, many applications have a mix of active and passive objects. They **create and destroy** objects.



OBJECT ORIENTED SOFTWARE ENGINEERING

2. Sequence diagram with Transient Object…

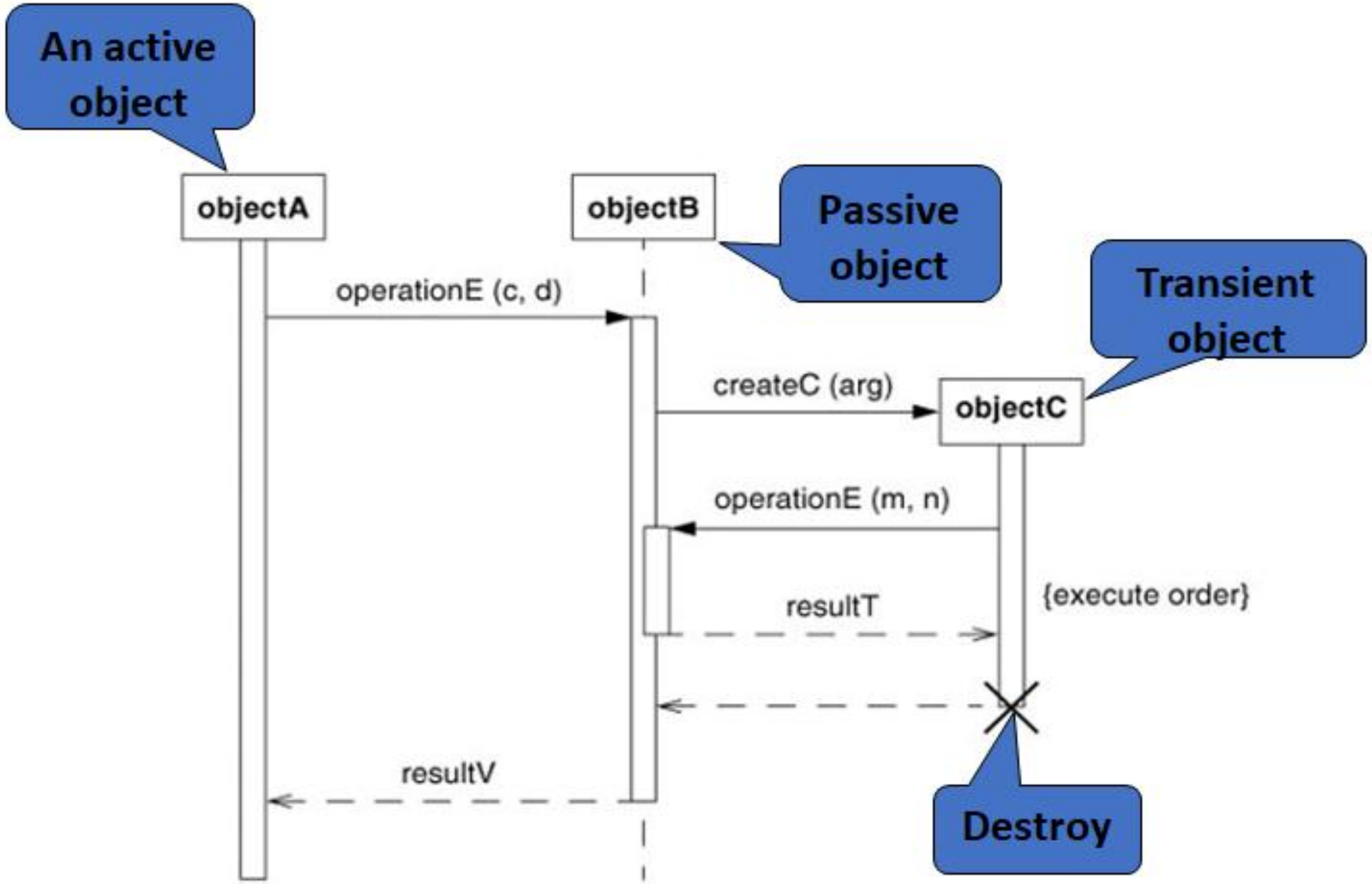


Figure Sequence diagram with a transient object. Many applications have a mix of active and passive objects. They create and destroy objects.

2. Sequence diagram with Transient Object…

- Here **object A** is an **active object** which initiates an operation, because it is active, its **activation rectangle spans the entire time** shown in diagram.

- **Object B** is a **passive object** that exists during the entire time shown in figure, but it is not active for the whole time.

- The UML shows its existence by dashed line Object B’s life line broadens into an activation rectangle when it is processing a call.

2. Sequence diagram with Transient Object…

- For some part of time, it performs a **recursive operation** shown by the **doubled activation** rectangle between the call by object C on operation E and returns the result values.

- Object C is created and destroyed during the time shown on diagram so its lifeline does not span the whole diagram.

Guidelines of Sequence diagram

1. **Prepare at least one scenario per use case:-** the steps in the scenario should be logical commands, not individual button clicks.
2. **Abstract the scenario into sequence diagrams:-**Sequence diagrams clearly shows the contribution of each actor.
3. **Divide complex interactions:-**Break large interaction into their constituent tasks and prepare a sequence diagram for each of them.
4. **Prepare a sequence diagram for error condition:-**shows the system response to error condition.

THANK YOU

Ms. Archana A
Department of Computer Applications

archana@pes.edu

+91 80 6666 3333 Extn 392